

A Statistical Analysis of Prize Money, Tournaments, and Players (1998–2025)

JiXianLi

Seokyoung Kim

Junbo Zhao

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1 Introduction

E-sports has evolved from a small group of enthusiasts' entertainment into a global industry over the past three decades. At the same time, the number of professional players, the frequency of the competition, and the number of players are continuously increasing, which leads to esports becoming a formal exercise. However, we still lack quantitative research on the long-term growth model of e-sports. For instance, over the past nearly three decades, how has the total prize money of e-sports changed over time? Are there significant differences in the distribution of individual player incomes between different years? And is there a clear correlation between the growth of active players and the total prize money of. In order to answer these questions, this report records the data of e-sports from 1998 to 2025, including total award money, total number of tournaments, mean and median tournament award pool, and mean and median earnings per player. This report's analysis part mainly includes three parts:

1. The trend of the award pool of the tournaments over the years.
2. The difference between the mean and the median, and what does that difference mean?
3. The relationship between the number of players and the award pool of the tournament?

2 Data Description

The data comes from esportsearnings.com, a community-maintained tracker that has been logging tournament results since the early 2000s. We pulled the year-level totals into one CSV (`Esports_Stats.csv`) covering 1998 through 2025, with one row per year. All dollar amounts are nominal USD, so part of the long-term growth is just inflation.

Table 1 lists the eight columns and what they mean.

Table 1: Variables in `Esports_Stats.csv` (1998-2025, 28 rows).

Variable	Type	Description
year	integer	Calendar year of the observation
prize_money	numeric (USD)	Sum of prize money awarded across all tracked tournaments that year
total_tournaments	integer	Number of tracked tournaments held that year
total_active_players	integer	Players with at least one tracked tournament result that year
mean_tournament_prize_pool	numeric (USD)	Average prize pool across all tournaments that year
median_tournament_prize_pool	numeric (USD)	Middle prize pool when tournaments are sorted by size
mean_earnings_per_player	numeric (USD)	Average winnings per active player that year
median_earnings_per_player	numeric (USD)	Middle player's winnings when players are sorted by total earnings

To make the rest of the report easier to follow, we grouped the 28 years into four eras using visible breakpoints in the data. The cutoffs are not arbitrary: total tournament count crossed 1,000 around 2010, hit roughly 6,000 around 2019, and dropped sharply after 2022. Table 2 summarizes each era.

Table 2: Esports industry by era. Total prize money is summed within each era; the other columns are within-era averages.

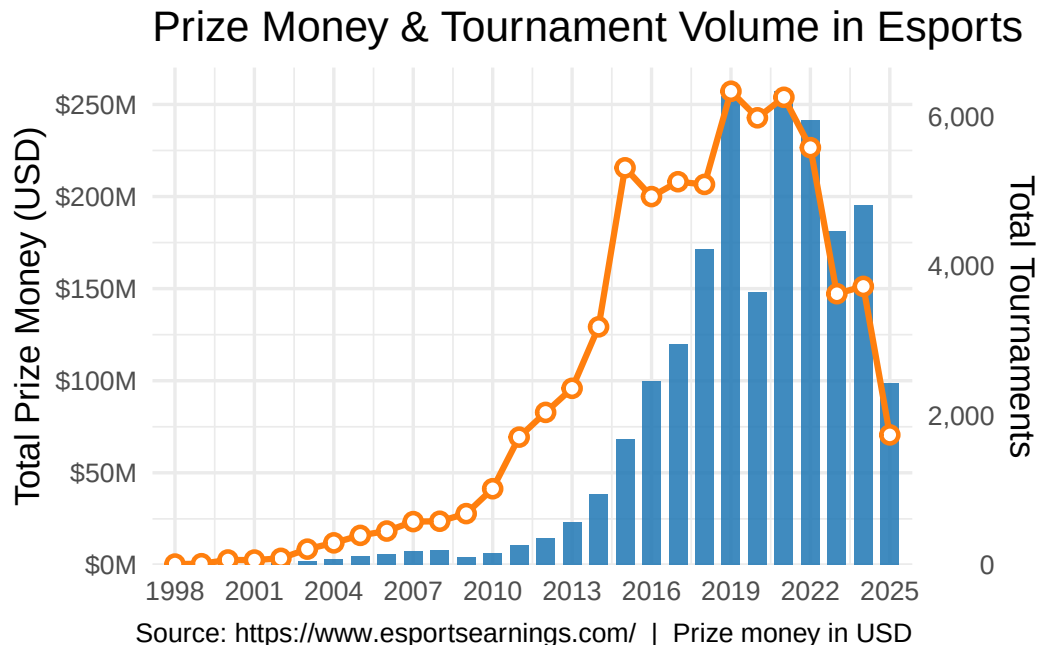
Era	Total prize money (USD)	Avg tournaments / yr	Avg active players / yr	Mean / median ratio
1. Early adoption (1998-2009)	37,910,835	287	1,330	3.5
2. Rapid growth (2010-2018)	551,047,449	3,418	11,499	24.8
3. Peak (2019-2022)	900,853,351	6,038	32,050	21.8
4. Contraction (2023-2025)	475,034,526	3,030	21,067	6.9

A few patterns are already visible from the table. The Rapid Growth era held more than ten times the tournaments per year that Early Adoption did, and the Peak era added another 75% on top of that. The mean-to-median ratio sat above 20 across both Rapid Growth and Peak, meaning a handful of mega-events were sitting on top of hundreds of much smaller ones. After 2022 that ratio collapsed into the single digits, but as Section 2.2 will show, the collapse was not because the small events finally caught up. It was because the small events stopped happening. The three analysis sections that follow walk through each piece of this picture in detail.

3 Data Analysis

3.1 The Relationship between the Total Prize Pool and The Number of Tournaments

Figure 1: Annual esports prize money (bars, left axis) and total tournaments held (line, right axis) from 1998 to 2025. Prize money values have been scaled to millions of USD for readability.



3.1.1 Analysis

This plot helps show how the overall scale of esports changed from 1998 to 2025 by comparing two measures at the same time: **total prize money** and **total tournaments**. The blue bars show how much prize money was awarded each year, while the orange line shows how many tournaments were held. Looking at both together makes it easier to see whether esports growth came from having **more events**, **larger prize pools**, or both.

There are three broad phases in the graph. **First**, from the late 1990s through about the early 2010s, both prize money and tournament counts were relatively low, though they gradually increased over time. **Second**, from about 2013 through 2019, esports expanded very quickly. Tournament counts rose sharply, and total prize money also rose at a fast rate. **Third**, the late 2010s to early 2020s appear to be the high point, with prize money peaking around 2021 and tournament counts remaining very high from about 2016 through 2022.

Since 2022, both metrics have trended downward, with 2025 data showing the lowest tournament count since 2013. This may reflect broader economic headwinds affecting esports investment, or a structural shift in how competitive gaming is organized and monetized.

A key insight from this graph is that **prize money and tournament volume do not always move at exactly the same rate**. For example, tournament counts remain high for several years

even when prize money fluctuates more strongly. That suggests the esports ecosystem can stay active even when the money side changes, and it may also show that some years had fewer or more concentrated high-prize events.

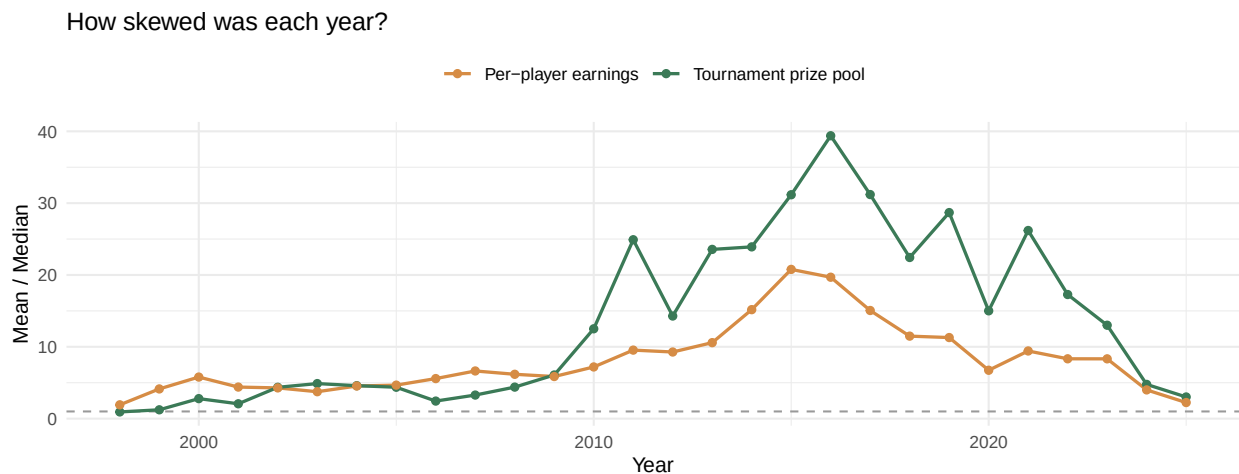
What this chart cannot tell us is whether the prize money was spread evenly across events or piled up in a few mega-tournaments. Section 2.2 takes a crack at exactly that question by comparing the mean and the median.

3.2 Mean vs. Median: How Skewed Are Esports Earnings?

Figure 2: Mean and median over time, on a log scale. Left: prize pool per tournament. Right: earnings per active player.



Figure 3: Ratio of mean to median over time. A value of 1 (dashed line) would mean a perfectly balanced distribution; everything above it is right-skew.



3.2.1 Analysis

We wanted to compare the mean and median here because they pick up on different things. The mean gets pulled around by outliers, one massive tournament like The International will push it way up even if most events that year were small. The median does not care about that at all; it just shows what the middle tournament looks like. So when these two numbers are far apart, it usually means the distribution is really skewed.

First, looking at Figure 2, we had to use a log scale because the range is kind of ridiculous. In 2010 the median tournament was paying around \$250, but the mean was already over \$6,000. On a normal axis you would not even be able to see the median line. From about 2010 to 2021, the two lines on the tournament side barely look related: the median stays somewhere in the low hundreds to low thousands while the mean is up in the tens of thousands. We plotted the actual ratio in Figure 3 to get a clearer picture, and for most of that stretch it is somewhere between 12x and 29x. Basically, for every big event like LoL Worlds there were a ton of small local tournaments that only paid out a couple hundred dollars, and those kept dragging the median down.

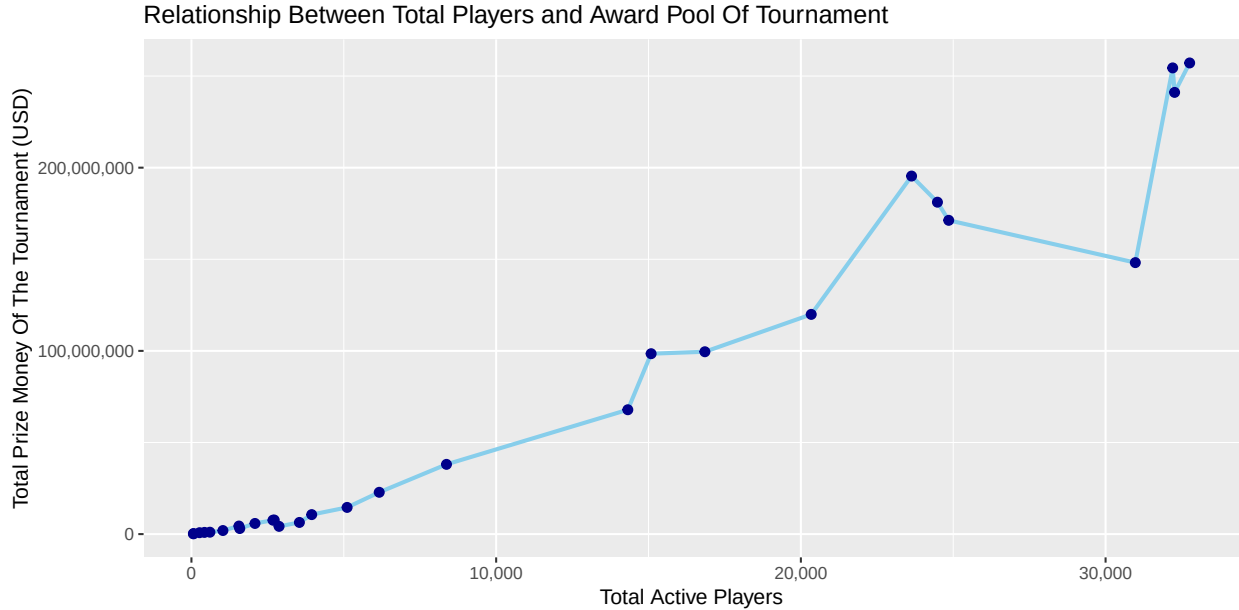
Second, on the player earnings side, the gap is there too but not as extreme. The mean is usually around 3 to 7 times the median, with a peak at about 15x in 2014. So a handful of top players were making a lot, but most people competing were earning way less than the average would suggest.

What caught our attention was the shift after 2022. The median tournament prize pool went from roughly \$1,500 in 2021 up to over \$18,000 by 2025, and the mean-to-median ratio dropped from around 26 all the way to about 3. At first that looks like a good thing, like the money is spreading out more evenly. But when we checked the total number of tournaments, the count went from about 6,300 in 2019 down to around 1,700 in 2025. So what actually happened is a lot of the smaller grassroots events stopped running. The tournaments that are left tend to be bigger and better funded, which is why the median jumped up and closed the gap with the mean. It is less that the distribution got more equal and more that the bottom end just went away.

So far we have looked at the totals (Section 2.1) and at how those totals were divided up within a year (Section 2.2). The last piece is the player side: when more people compete, does the pot grow with them?

3.3 The Relationship Between The Number Of Players And The Award Pool Of The Tournament

Figure 4: Relationship Between Total Players and Award Pool Of Tournament



3.3.1 Analysis

As the Figure 4 shows, overall, there is a strong positive correlation. That is, the more active players there are, the higher the total prize money for the event. This is intuitive, as an increase in participants usually means an expansion of the market size, which attracts more sponsors, viewers, and capital inflows, thereby raising the prize pool. Moreover, this line is not a uniform straight line; it can be split into three parts. First is the player number from 0 to 10000; in this case, the growth of bonuses has been very slow, which is in line with the early stage of the e-sports industry. At that time, the player base size was small, and the commercialization level was low. Second, is player number from 10001 to 22000, in this case, the slope has significantly steepened, and the bonus amount has rapidly increased from several hundred million dollars to nearly 200 million dollars. This indicates that within this range, the increase in bonus for each additional active player is much higher than in the early stages. Finally, is player number greater than 22001. An interesting phenomenon emerges here: the number of players decreases first and then rises sharply. This indicates that the relationship between the number of players and the prize money is no longer strictly linear. In some years, there are more players, but the prize money is actually lower. Although there is a strong positive correlation shown in the graph, it cannot be simply concluded that “more players lead to more bonuses”. Because supply and demand are mutually restrictive: as the bonus pool expands, it attracts more players to join, while an increase in the player base in turn attracts sponsors to make further investments, forming a positive feedback loop.

4 Summary

All three sections were aimed at the same bigger question: how did esports actually grow, and what is happening now that the growth has stopped. Read together, they tell one story.

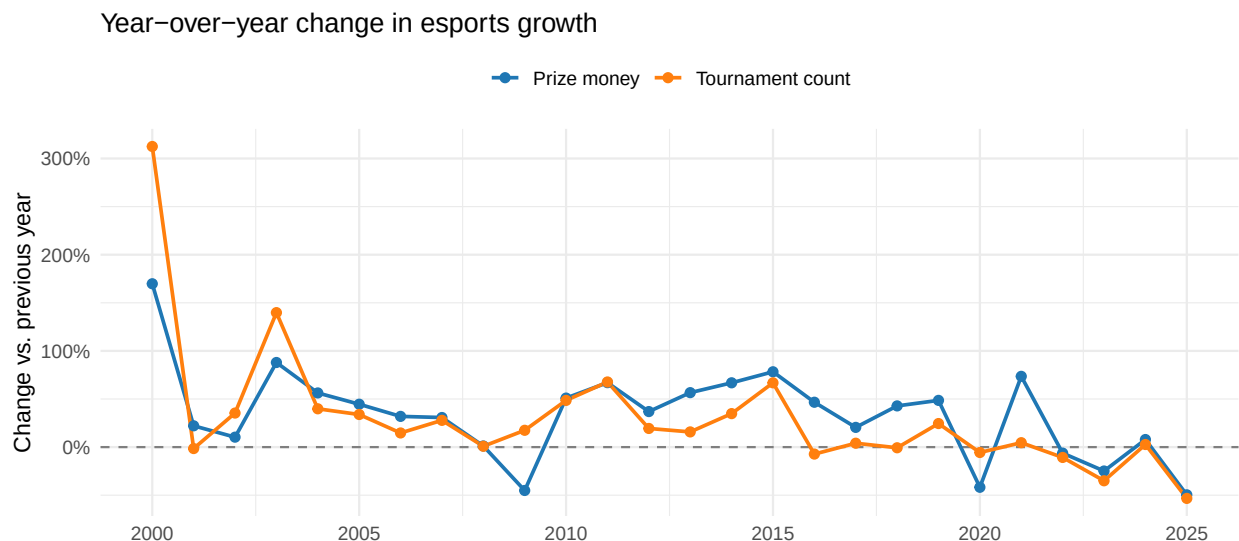
Section 2.1 showed that total prize money and tournament count rose roughly together until about 2019, when the tournament count peaked at around 6,300 and started falling. By 2025 the count was back down to about 1,700. Prize money stayed elevated through 2021 and only dropped sharply after 2022. The two lines stopped tracking each other.

Section 2.2 zoomed into what that meant for a typical event. From 2010 to 2021 the mean tournament prize pool ran 12 to 29 times the median, which is a textbook right skew: a handful of mega-events sitting on top of hundreds of small ones. After 2022 the ratio collapsed to about 3, but the closing of that gap was not because the small events caught up. It was because the small events stopped happening.

Section 2.3 showed that more active players generally meant more prize money, but the curve loops back on itself across 2020 and 2021, when player counts stayed high while in-person events could not run.

Figure 5 puts the timing on a single chart. Both metrics drop hard in 2020, recover unevenly through 2021, then both turn negative from 2023 onward. The 2020 dip is short and tied to a single global event. The slower bleed after 2022 is harder to explain away, since it is not tied to a single shock and it shows up in both prize money and tournament count at the same time.

Figure 5: Year-over-year percent change in total prize money and tournament count, 2000-2025. Values above the dashed line indicate growth from the previous year; values below indicate contraction.



4.1 Limitations

A few things this dataset cannot tell us. It does not split tournaments by game title, so Dota 2's The International gets averaged in with small Counter-Strike LANs. It only counts players who have at least one recorded prize, so casual or unsponsored competitors are invisible to us. The dollar figures are nominal, so part of the long-term growth is just inflation. The 2025 numbers are likely incomplete too, since esportsearnings.com keeps updating older years as results come in.

5 Appendix

5.1 Appendix A: GenAI Usage

This appendix documents all uses of generative AI tools in this project, following the course academic integrity policy.

5.1.1 Usage 1: Introduction Section Reference For Data analysis QMD file

- **Tool:** Claude 4.7 adaptive(Anthropic)
- **Date:** April 24, 2026
- **Purpose:** Reference for what topics belong in the Introduction of a data analysis report. Claude's output served as a structural checklist (context, research question, scope, data sources).
- **Exact prompt used:**

What should be included in the introduction section of a data analysis report?

- **Exact response generated by Claude:** Start with the context and background — what situation, problem, or domain prompted this analysis? Then state the purpose and objectives clearly. What question are you trying to answer, or what decision is this analysis meant to support? Briefly describe the scope — what's included and, just as usefully, what isn't. Time periods, populations, geographies, product lines, any boundaries that shape how the findings should be read. Mention the data sources at a high level (full details go in a methodology section).

5.1.2 Usage 2: Long-Description Reference for Figure 5

- **Tool:** Claude Opus 4.7 (Anthropic)
- **Date:** May 4, 2026
- **Purpose:** Reference draft for the accessibility long description (**fig-alt**) of Figure 5, the year-over-year growth chart in the Summary section. Claude’s output was used as a starting reference for tone and level of detail; the **fig-alt** text that appears in the final report was rewritten in our own words.
- **Exact prompt used:**

I have a line chart that compares the year-over-year percent change of two esports metrics, total prize money and total tournament count, from 2000 through 2025. I have already upload the chart photo. Can you write a 2 to 3 sentence accessibility long description for this chart? Describe what the axes show and the overall shape of the trends. Do not include any wording from a course assignment.

- **Exact response generated by Claude:** Claude returned a three-sentence long description. It named the two lines (prize money and tournament count), pointed out the dashed reference line at zero, and summarized the shape of the curves: a steep drop in 2020 followed by an uneven recovery in 2021, and a return to negative growth from 2023 through 2025. The draft also noted that the two metrics had moved together earlier in the range but diverged after 2022.
- **How the response was used:** After reading Claude’s draft, we rewrote the long description in our own words and adjusted phrasing so the final **fig-alt** text does not match the draft sentence-by-sentence. The wording that appears in the QMD chunk for Figure 5 is our own.

5.1.3 Note on Grammarly (spelling and grammar only)

Grammarly (browser extension) was used for grammar and spelling correction on text that the team had already written ourselves. Under the course policy, spelling/grammar tools are exempt from formal disclosure. We note it here only for transparency. Grammarly does not accept natural-language prompts; its output is limited to inline word- and clause-level corrections, each of which was reviewed and accepted or rejected manually. No portion of the report’s writing was generated by Grammarly.

5.1.4 Authors

Seokyoung Kim, Junbo Zhao, JiXian Li

5.2 Appendix B: Code Appendix

The chunk below auto-includes every code chunk from the body so we only have to maintain one copy of each plot or table.

```
esports <- read_csv("Esports_Stats.csv", show_col_types = FALSE) |>
  rename(
    prize_money          = `Total Prize Money`,
    total_tournaments    = `Total Tournaments`,
    total_active_players = `Total Active Players`,
    mean_tournament_prize_pool = `Mean Tournament Prize Pool`,
    mean_earnings_per_player = `Mean Earnings/Player`,
    median_tournament_prize_pool = `Median Tournament Prize Pool`,
    median_earnings_per_player = `Median Earnings/Player`
  )
tibble(
  Variable = c("year", "prize_money", "total_tournaments",
               "total_active_players",
               "mean_tournament_prize_pool",
               "median_tournament_prize_pool",
               "mean_earnings_per_player",
               "median_earnings_per_player"),
  Type = c("integer", "numeric (USD)", "integer", "integer",
           "numeric (USD)", "numeric (USD)",
           "numeric (USD)", "numeric (USD)"),
  Description = c(
    "Calendar year of the observation",
    "Sum of prize money awarded across all tracked tournaments that year",
    "Number of tracked tournaments held that year",
    "Players with at least one tracked tournament result that year",
    "Average prize pool across all tournaments that year",
    "Middle prize pool when tournaments are sorted by size",
    "Average winnings per active player that year",
    "Middle player's winnings when players are sorted by total earnings"
  )
) |>
  knitr::kable()
esports |>
  mutate(era = case_when(
    year <= 2009 ~ "1. Early adoption (1998-2009)",
    year <= 2018 ~ "2. Rapid growth (2010-2018)",
    year <= 2022 ~ "3. Peak (2019-2022)",
    TRUE        ~ "4. Contraction (2023-2025)"
  )) |>
  group_by(era) |>
  summarize(
    total_prize_usd = round(sum(prize_money)),
    avg_tournaments = round(mean(total_tournaments)),
```

```

    avg_players      = round(mean(total_active_players)),
    avg_mm_ratio     = round(mean(mean_tournament_prize_pool /
                                  median_tournament_prize_pool), 1),

    .groups = "drop"
  ) |>
  knitr::kable(
    col.names = c("Era", "Total prize money (USD)",
                  "Avg tournaments / yr", "Avg active players / yr",
                  "Mean / median ratio"),
    format.args = list(big.mark = ",")
  )
df <- esports |>
  select(year, prize_money, total_tournaments)

scale_factor <- max(df$prize_money) / max(df$total_tournaments)

col_prize <- "#1f77b4"  # prize money
col_tourn <- "#ff7f0e"  # tournament count

p <- ggplot(df, aes(x = year)) +

  geom_col(
    aes(y = prize_money),
    fill = col_prize,
    alpha = 0.85,
    width = 0.7
  ) +
  geom_line(
    aes(y = total_tournaments * scale_factor),
    color = col_tourn,
    linewidth = 1.1
  ) +
  geom_point(
    aes(y = total_tournaments * scale_factor),
    color = col_tourn,
    size = 2.2,
    shape = 21,
    fill = "white",
    stroke = 1.2
  ) +

  # Dual y-axes
  scale_y_continuous(
    name = "Total Prize Money (USD)",
    labels = function(x) paste0("$", x / 1e6, "M"),
    expand = expansion(mult = c(0, 0.05)),
    sec.axis = sec_axis(

```

```

      transform = ~ . / scale_factor,
      name      = "Total Tournaments",
      labels    = function(x) formatC(x, format = "d", big.mark = ",")
    )
  ) +
  scale_x_continuous(
    breaks = seq(1998, 2025, by = 3),
    expand = expansion(add = 0.5)
  ) +

  theme_minimal(base_size = 13) +

  labs(
    title    = "Prize Money & Tournament Volume in Esports",
    x        = NULL,
    caption  = "Source: https://www.esportsearnings.com/ | Prize money in USD"
  )
p

mvm_long <- esports |>
  select(year,
    mean_tournament_prize_pool, median_tournament_prize_pool,
    mean_earnings_per_player, median_earnings_per_player) |>
  pivot_longer(-year, names_to = "metric", values_to = "value") |>
  mutate(
    panel = if_else(str_detect(metric, "tournament"),
      "Per tournament prize pool",
      "Per player earnings"),
    center_type = if_else(str_detect(metric, "^mean"), "Mean", "Median")
  )

ggplot(mvm_long, aes(x = year, y = value, color = center_type)) +
  geom_line(linewidth = 0.8) +
  geom_point(size = 1.6) +
  facet_wrap(~ panel, scales = "free_y") +
  scale_y_log10(labels = dollar) +
  scale_color_manual(values = c("Mean" = "#c0392b", "Median" = "#2c3e50")) +
  labs(
    title = "Where the average meets the typical",
    x = "Year", y = "USD (log scale)", color = NULL
  ) +
  theme_minimal(base_size = 11) +
  theme(legend.position = "top",
    strip.text = element_text(face = "bold"))
ratio_long <- esports |>
  transmute(
    year,

```



```

`Tournament prize pool` =
  mean_tournament_prize_pool / median_tournament_prize_pool,
`Per-player earnings` =
  mean_earnings_per_player / median_earnings_per_player
) |>
pivot_longer(-year, names_to = "metric", values_to = "ratio")

ggplot(ratio_long, aes(x = year, y = ratio, color = metric)) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "grey60") +
  geom_line(linewidth = 0.8) +
  geom_point(size = 1.6) +
  scale_color_manual(values = c("Tournament prize pool" = "#3b7a57",
                                "Per-player earnings" = "#d68c45")) +
  labs(
    title = "How skewed was each year?",
    x = "Year", y = "Mean / Median", color = NULL
  ) +
  theme_minimal(base_size = 11) +
  theme(legend.position = "top")
ggplot(esports, aes(x = total_active_players, y = prize_money)) +
  geom_line(color = "skyblue", linewidth = 1) +
  geom_point(color = "darkblue", size = 2) +
  scale_x_continuous(labels = comma) +
  scale_y_continuous(labels = comma) +
  labs(
    title = "Relationship Between Total Players and Award Pool Of Tournament",
    x = "Total Active Players",
    y = "Total Prize Money Of The Tournament (USD)"
  )
esports |>
  arrange(year) |>
  mutate(
    prize_yoy = (prize_money - lag(prize_money)) /
      lag(prize_money) * 100,
    tourn_yoy = (total_tournaments - lag(total_tournaments)) /
      lag(total_tournaments) * 100
  ) |>
  filter(year >= 2000) |>
  pivot_longer(c(prize_yoy, tourn_yoy),
    names_to = "metric", values_to = "yoy_pct") |>
  mutate(metric = recode(metric,
    prize_yoy = "Prize money",
    tourn_yoy = "Tournament count")) |>
  ggplot(aes(x = year, y = yoy_pct, color = metric)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey50") +
  geom_line(linewidth = 0.9) +
  geom_point(size = 2) +

```

```

scale_color_manual(values = c("Prize money"      = "#1f77b4",
                              "Tournament count" = "#ff7f0e")) +
scale_y_continuous(labels = function(x) paste0(x, "%")) +
scale_x_continuous(breaks = seq(2000, 2025, by = 5)) +
labs(
  title = "Year-over-year change in esports growth",
  x = NULL, y = "Change vs. previous year", color = NULL
) +
theme_minimal(base_size = 12) +
theme(legend.position = "top")

```